



# Mechanical Ventilation; Adult (Optional)

## History

- Multiple etiologies leading to need for advanced airway control
- Requires ventilation support
- Height and underlying lung conditions

## Signs and Symptoms

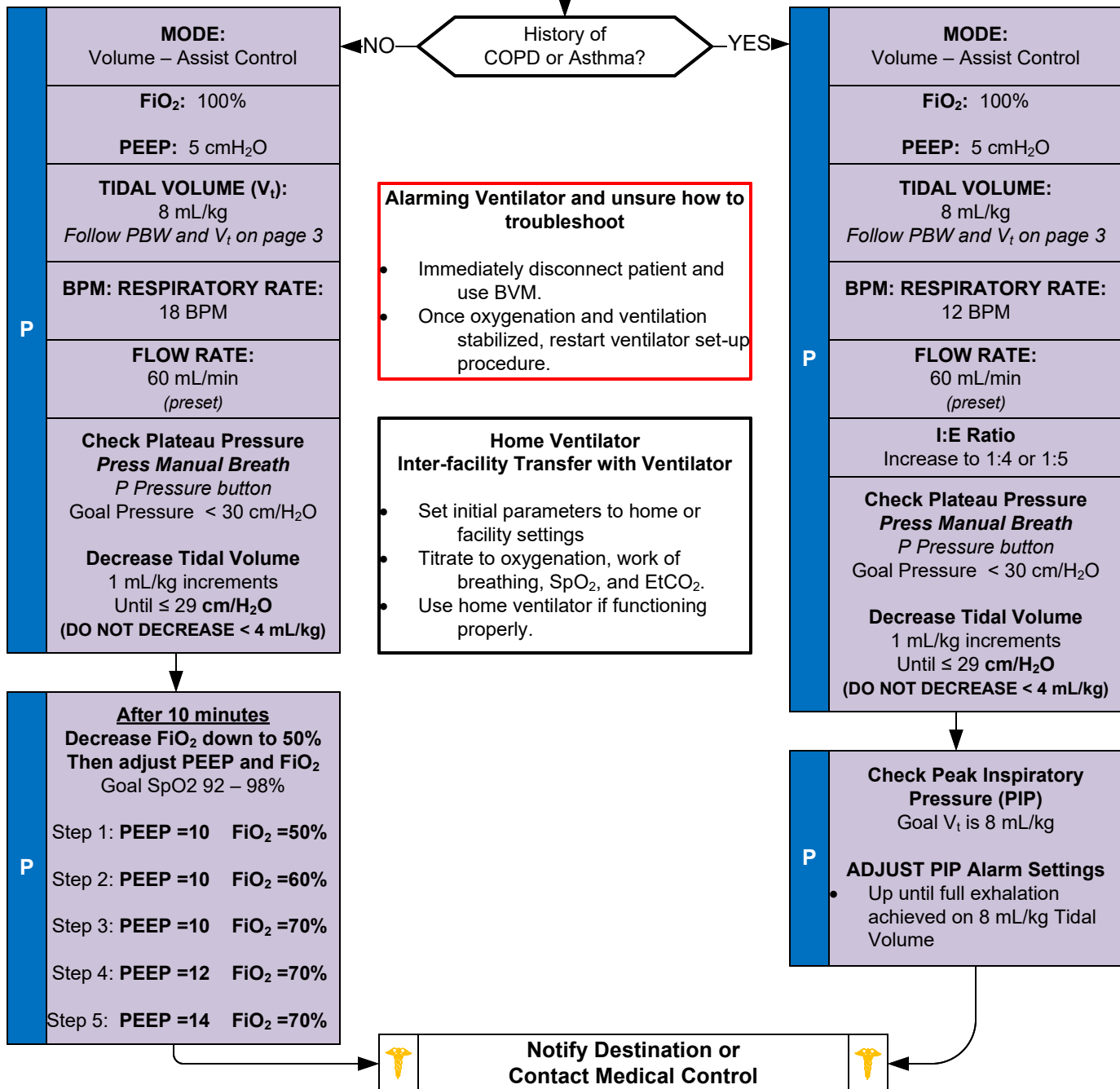
- Loss of consciousness or AMS with inability to protect airway
- Difficult oxygenation and/or ventilation
- 

## Differential

- ROSC
- Trauma
- Stroke
- Seizure
- Shock (see Shock Protocol)
- Toxicological

Age Appropriate  
 Airway Protocol(s) AR 1, 2, 3, 5, 6  
**if indicated**

Post-intubation/BIAD Management  
 Protocol AR 8  
**if indicated**





# Mechanical Ventilation; Adult (Optional)

## Pearls

- **Recommended Exam: Mental Status, Skin, HEENT, Heart, Lungs, Abdomen, Back, Extremities, Neuro**
- **Mechanical ventilation may be used in any patient  $\geq 1$  year old.**
- **MODE:**
  - In all adult patients use Volume – Assist Control.
  - This mode requires adequate sedation as it can be uncomfortable in a patient who is awakening.
- **TIDAL VOLUME:**
  - Tidal volume is very important in preventing lung injury and calculated by height and predicted body weight, or ideal body weight, and NOT actual body weight.
  - Follow Tidal Volume by Height Table on page 3.
  - Follow Tidal Volume by Height Table on page 3 when adjusting Peak Inspiratory Pressure alarms to allow full exhalation.
  - High Tidal Volumes are well known to cause alveolar damage and lung injury.
- **FLOW RATE:**
  - A normal breath (non-mechanical ventilation) has highest flow and volume at the beginning and both decrease as inspiration comes to an end.
  - Setting Flow Rate at 60 L/minute allows patient to take full breath without air hunger toward end of inspiration. This is more comfortable for the patient.
  - If patient looks like they are trying to take in more volume initially, the Flow Rate can be increased by increments of 5 as needed to improve patient comfort.
- **FiO<sub>2</sub> and PEEP Adjustments:**
  - Seems intuitive that when SpO<sub>2</sub> is less than desired the FiO<sub>2</sub> should be increased.
  - When FiO<sub>2</sub> is  $\geq 50\%$  and SpO<sub>2</sub> remains low, this indicates a shunt, and PEEP must be used in conjunction with FiO<sub>2</sub> to correct the shunt and increase oxygenation.
  - Follow PEEP adjustment recommendations on page 1.
- **EtCO<sub>2</sub>:**
  - EtCO<sub>2</sub> and arterial CO<sub>2</sub> do not always correlate well in patients with lung disease or during serious illness or injury.
  - Use caution in adjusting respiratory rate to reach a goal of 35 – 45 mmHg. Most intubated patients do not need tight control in this range.
  - Patients with suspected head injury do need EtCO<sub>2</sub> with a target of 35 – 45 mmHg.
  - Allowing patients with COPD and asthma exacerbations to have higher EtCO<sub>2</sub> outside the 35 – 45 mmHg range is acceptable. Lower ventilation rates allow more time for exhalation and prevents auto-PEEP and/ or air trapping.
- **DOPE:** Displaced tracheostomy tube / ETT, Obstructed tracheostomy tube / ETT, Pneumothorax and Equipment failure.



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## TIDAL VOLUME INITIAL SETTINGS By HEIGHT

### FEMALE

Height / Predicted body weight / Vt

| HEIGHT      | PBW   | 4 ml | 5 ml | 6 ml | 7 ml | 8 ml |
|-------------|-------|------|------|------|------|------|
| 4' 0" (48)  | 17.9  | 72   | 90   | 107  | 125  | 143  |
| 4' 1" (49)  | 20.2  | 81   | 101  | 121  | 141  | 162  |
| 4' 2" (50)  | 22.5  | 90   | 113  | 135  | 158  | 180  |
| 4' 3" (51)  | 24.8  | 99   | 124  | 149  | 174  | 198  |
| 4' 4" (52)  | 27.1  | 108  | 136  | 163  | 190  | 217  |
| 4' 5" (53)  | 29.4  | 118  | 147  | 176  | 206  | 235  |
| 4' 6" (54)  | 31.7  | 127  | 159  | 190  | 222  | 254  |
| 4' 7" (55)  | 34    | 136  | 170  | 204  | 238  | 272  |
| 4' 8" (56)  | 36.3  | 145  | 182  | 218  | 254  | 290  |
| 4' 9" (57)  | 38.6  | 154  | 193  | 232  | 270  | 309  |
| 4' 10" (58) | 40.9  | 164  | 205  | 245  | 286  | 327  |
| 4' 11" (59) | 43.2  | 173  | 216  | 259  | 302  | 346  |
| 5' 0" (60)  | 45.5  | 182  | 228  | 273  | 319  | 364  |
| 5' 1" (61)  | 47.8  | 191  | 239  | 287  | 335  | 382  |
| 5' 2" (62)  | 50.1  | 200  | 251  | 301  | 351  | 401  |
| 5' 3" (63)  | 52.4  | 210  | 262  | 314  | 367  | 419  |
| 5' 4" (64)  | 54.7  | 219  | 274  | 328  | 383  | 438  |
| 5' 5" (65)  | 57    | 228  | 285  | 342  | 399  | 456  |
| 5' 6" (66)  | 59.3  | 237  | 297  | 356  | 415  | 474  |
| 5' 7" (67)  | 61.6  | 246  | 308  | 370  | 431  | 493  |
| 5' 8" (68)  | 63.9  | 256  | 320  | 383  | 447  | 511  |
| 5' 9" (69)  | 66.2  | 265  | 331  | 397  | 463  | 530  |
| 5' 10" (70) | 68.5  | 274  | 343  | 411  | 480  | 548  |
| 5' 11" (71) | 70.8  | 283  | 354  | 425  | 496  | 566  |
| 6' 0" (72)  | 73.1  | 292  | 366  | 439  | 512  | 585  |
| 6' 1" (73)  | 75.4  | 302  | 377  | 452  | 528  | 603  |
| 6' 2" (74)  | 77.7  | 311  | 389  | 466  | 544  | 622  |
| 6' 3" (75)  | 80    | 320  | 400  | 480  | 560  | 640  |
| 6' 4" (76)  | 82.3  | 329  | 412  | 494  | 576  | 658  |
| 6' 5" (77)  | 84.6  | 338  | 423  | 508  | 592  | 677  |
| 6' 6" (78)  | 86.9  | 348  | 435  | 521  | 608  | 695  |
| 6' 7" (79)  | 89.2  | 357  | 446  | 535  | 624  | 714  |
| 6' 8" (80)  | 91.5  | 366  | 458  | 549  | 641  | 732  |
| 6' 9" (81)  | 93.8  | 375  | 469  | 563  | 657  | 750  |
| 6' 10" (82) | 96.1  | 384  | 481  | 577  | 673  | 769  |
| 6' 11" (83) | 98.4  | 394  | 492  | 590  | 689  | 787  |
| 7' 0" (84)  | 100.7 | 403  | 504  | 604  | 705  | 806  |

### MALE

Height / Predicted body weight / Vt

| HEIGHT      | PBW   | 4 ml | 5 ml | 6 ml | 7 ml | 8 ml |
|-------------|-------|------|------|------|------|------|
| 4' 0" (48)  | 22.4  | 90   | 112  | 134  | 157  | 179  |
| 4' 1" (49)  | 24.7  | 99   | 124  | 148  | 173  | 198  |
| 4' 2" (50)  | 27    | 108  | 135  | 162  | 189  | 216  |
| 4' 3" (51)  | 29.3  | 117  | 147  | 176  | 205  | 234  |
| 4' 4" (52)  | 31.6  | 126  | 158  | 190  | 221  | 253  |
| 4' 5" (53)  | 33.9  | 136  | 170  | 203  | 237  | 271  |
| 4' 6" (54)  | 36.2  | 145  | 181  | 217  | 253  | 290  |
| 4' 7" (55)  | 38.5  | 154  | 193  | 231  | 270  | 308  |
| 4' 8" (56)  | 40.8  | 163  | 204  | 245  | 286  | 326  |
| 4' 9" (57)  | 43.1  | 172  | 216  | 259  | 302  | 345  |
| 4' 10" (58) | 45.4  | 182  | 227  | 272  | 318  | 363  |
| 4' 11" (59) | 47.7  | 191  | 239  | 286  | 334  | 382  |
| 5' 0" (60)  | 50    | 200  | 250  | 300  | 350  | 400  |
| 5' 1" (61)  | 52.3  | 209  | 262  | 314  | 366  | 418  |
| 5' 2" (62)  | 54.6  | 218  | 273  | 328  | 382  | 437  |
| 5' 3" (63)  | 56.9  | 228  | 285  | 341  | 398  | 455  |
| 5' 4" (64)  | 59.2  | 237  | 296  | 355  | 414  | 474  |
| 5' 5" (65)  | 61.5  | 246  | 308  | 369  | 431  | 492  |
| 5' 6" (66)  | 63.8  | 255  | 319  | 383  | 447  | 510  |
| 5' 7" (67)  | 66.1  | 264  | 331  | 397  | 463  | 529  |
| 5' 8" (68)  | 68.4  | 274  | 342  | 410  | 479  | 547  |
| 5' 9" (69)  | 70.7  | 283  | 354  | 424  | 495  | 566  |
| 5' 10" (70) | 73    | 292  | 365  | 438  | 511  | 584  |
| 5' 11" (71) | 75.3  | 301  | 377  | 452  | 527  | 602  |
| 6' 0" (72)  | 77.6  | 310  | 388  | 466  | 543  | 621  |
| 6' 1" (73)  | 79.9  | 320  | 400  | 479  | 559  | 639  |
| 6' 2" (74)  | 82.2  | 329  | 411  | 493  | 575  | 658  |
| 6' 3" (75)  | 84.5  | 338  | 423  | 507  | 592  | 676  |
| 6' 4" (76)  | 86.8  | 347  | 434  | 521  | 608  | 694  |
| 6' 5" (77)  | 89.1  | 356  | 446  | 535  | 624  | 713  |
| 6' 6" (78)  | 91.4  | 366  | 457  | 548  | 640  | 731  |
| 6' 7" (79)  | 93.7  | 375  | 469  | 562  | 656  | 750  |
| 6' 8" (80)  | 96    | 384  | 480  | 576  | 672  | 768  |
| 6' 9" (81)  | 98.3  | 393  | 492  | 590  | 688  | 786  |
| 6' 10" (82) | 100.6 | 402  | 503  | 604  | 704  | 805  |
| 6' 11" (83) | 102.9 | 412  | 515  | 617  | 720  | 823  |
| 7' 0" (84)  | 105.2 | 421  | 526  | 631  | 736  | 842  |

### TROUBLESHOOTING Hypoxia or Deterioration DOPEs

### RESPONSE to Hypoxia or Deterioration DOTT

|          |                                                                     |          |                                                                               |
|----------|---------------------------------------------------------------------|----------|-------------------------------------------------------------------------------|
| <b>D</b> | Dislodged ETT or cuff leak                                          | <b>D</b> | Disconnect ventilator, squeeze chest if auto-PEEP, Decompress if pneumothorax |
| <b>O</b> | Obstruction of ETT or circuit                                       | <b>O</b> | Oxygen 100% FiO2, BVM and check compliance                                    |
| <b>P</b> | Pneumothorax, Pneumonia, Pulmonary embolism or edema, Plug (mucous) | <b>T</b> | Tube position and function, check EtCO2                                       |
| <b>E</b> | Equipment problem                                                   | <b>T</b> | Tweak ventilator settings or equipment                                        |
| <b>S</b> | Stacked breaths, air trapping, or auto-PEEP                         |          |                                                                               |

| Pressure Alarm Troubleshooting |   |                     | Problem Location | Consider                                                                                                                                                         |
|--------------------------------|---|---------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| High PIP                       | + | High Plateau > 30   | Alveoli          | Compliance problem: Pneumothorax, Pneumonia<br>Pulmonary Edema or Embolism, CHF                                                                                  |
| High PIP                       | + | Normal Plateau < 30 | Airway problem   | Airway, ventilator, or circuit problem: DOPE, Right Main<br>stem intubation, Air trapping or auto-PEEP, Mucous<br>plug, Patient out of synchrony with ventilator |